Coding & Solution Summary

Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Coding & Solution Summary
Project Phase	Phase 5: Project Development Phase
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Organization / College	Vishnu Institute of Technology
Status / Version	Approved / v1.0 (Exact Replica Format)

1. Deployment & Code Parameters

Parameter	Target Repository Details	Compliance Notes
Repository URL	https://github.com/dharani756/Smart-Lender	Main Git Repository
Framework Environment	Python v3.10+ & Flask v3.0+	Applies light WSGI microserver configuration
Local Development Domain	http://127.0.0.1:5000 (Localhost Port 5000)	Server startup URL

2. Server Integration Snippet

```python

    # predictions using the loaded model file
    prediction=model.predict(data)
    print(prediction)
    prediction = int(prediction)
    print(type(prediction))

    if (prediction == 0):
        return render_template("output.html",result ="Loan wiil Not be Approved")
    else:
        return render_template("output.html",result = "Loan will be Approved")
    # showing the prediction results in a UI
if __name__=="__main__":

    # app.run(host='0.0.0.0', port=8000,debug=True)    # running the app
    port=int(os.environ.get('PORT',5000))
    app.run(debug=False)
```

Figure: Flask app.py Route Mapping and Prediction Operations

3. Setup Checklists

- Step 1: Clone the repository to the local system folder: git clone https://github.com/dharani756/Smart-Lender
- Step 2: Set up environment and install required dependencies: pip install -r requirements.txt
- Step 3: Run the web endpoint launcher: python app.py